

# Fiber Optics

December 6, 2025

## **Abstract**

‘Fiber Optics’ class, 1<sup>st</sup> semester, A.Y. 2025/26

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# 1 Lecture - Oct. 9

## 1.1 Maxwell's equations

### 1.1.1 Time Domain

$$\vec{\nabla} \times \vec{e} = -\frac{\partial \vec{b}}{\partial t} \quad \vec{\nabla} \times \vec{h} = \vec{j} + \frac{\partial \vec{d}}{\partial t} \quad \vec{\nabla} \cdot \vec{d} = \rho \quad \vec{\nabla} \cdot \vec{b} = 0$$

We'll assume  $\rho = 0$  and also  $\vec{j} = 0$

### 1.1.2 Harmonic Domain

We know that  $\vec{e}(t) = \Re [\vec{E} e^{j\omega t}] = \dots = \vec{E}_{Re} \cos(\omega t) - \vec{E}_{Im} \sin(\omega t)$ , hence:

$$\vec{\nabla} \times \vec{E} = -j\omega \vec{B} \quad \vec{\nabla} \times \vec{H} = j\omega \vec{D} \quad \vec{\nabla} \cdot \vec{D} = 0 \quad \vec{\nabla} \cdot \vec{B} = 0$$

Relationships between  $\vec{E}, \vec{H}$  to  $\vec{B}, \vec{D}$  goes through  $\epsilon, \mu$ : values are not always constant, and can cause dispersion.

For dielectric material:  $\vec{B} = \mu \vec{H}$

For isotropic material:  $\vec{D} = \epsilon \vec{E}$

Especially  $\vec{D}$  can vary with  $\omega$ , so also  $\epsilon$  vary (source of distortion).

## 1.2 Helmholtz's Equations

### 1.2.1 Harmonic Domain

$$\vec{\nabla} \times \vec{E} = -j\omega \mu_0 \vec{H} \quad \vec{\nabla} \times \vec{H} = j\omega \epsilon \vec{E} \quad \vec{\nabla} \cdot \vec{D} = 0 \quad \vec{\nabla} \cdot \vec{B} = 0$$

Trick: dividing space in regions where  $\epsilon$  is constant, we can simplify equations and get Laplacian:

$$\nabla^2 \vec{\square} + k^2 \vec{\square} = 0 \quad k^2 = \omega^2 \mu_0 \epsilon \quad \square \in \{\vec{E}, \vec{H}\}$$

We get N different solutions with N as number of regions as defined before.

### 1.2.2 Boundary Conditions

Consider the thin surface between 2 different materials. We define the normal vector  $\vec{U}_n$ .

For flux densities there are 2 bound conditions (continuity in **normal** component):

- $(\vec{B}_2 - \vec{B}_1) \cdot \vec{U}_n = 0$
- $(\vec{D}_2 - \vec{D}_1) \cdot \vec{U}_n = \rho_s$  and for our assumption  $\rho_s = 0$

For fields there are another 2 bound conditions (continuity in **tangent** component):

- $(\vec{E}_2 - \vec{E}_1) \times \vec{U}_n = 0$
- $(\vec{H}_2 - \vec{H}_1) \times \vec{U}_n = \vec{j}_s$ , also for our assumption  $= 0$

## 1.3 Poynting Vector

### 1.3.1 Time domain formulation

$$\vec{p}(t) = \vec{e}(t) \times \vec{h}(t)$$

### 1.3.2 Harmonic domain formulation

$$\vec{P} = \frac{1}{2} \vec{E} \times \vec{H}^* \quad \left[ \frac{W}{m^2} \right]$$

## 1.4 Complex Envelope

We know that  $\vec{e}(t) \xrightarrow{\text{Fourier}} \hat{e}(f) \xrightarrow{\text{Herm.Symm.}} \hat{e}^*(-f)$

We take our signal and we subtract the carrier frequency with the  $e^{-j2\pi f_c t}$  term, such that:

$$\bar{E}(t) = e^{-j2\pi f_0 t} \cdot \vec{e}^{(a)}(t) \quad \implies \quad \nabla^2 \bar{E}(\omega) + k^2 \bar{E}(\omega) = 0$$

## 2 Lecture - Oct. 10

### 2.1 Polarization

At a given time instant, the field could be written as:

$$\vec{e}(t) = \sum_{n=1}^{3dim.} \vec{U}_n c_n \cos(\omega t + \phi_n)$$

after cosine decomposition we can find:

$$\vec{e}(t) = \dots = \vec{c} \cos(\omega t) - \vec{c}' \sin(\omega t) \quad \text{with} \quad \begin{cases} \vec{c} = \sum_{n=1}^{3dim.} \vec{U}_n c_n \cos(\phi_n) \\ \vec{c}' = \sum_{n=1}^{3dim.} \vec{U}_n c_n \sin(\phi_n) \end{cases}$$

From previous lecture (Maxwell in Harmonic Domain), we know that

$$\vec{e}(t) = \Re [\vec{E} e^{j\omega t}] = \dots = \vec{E}_{Re} \cos(\omega t) - \vec{E}_{Im} \sin(\omega t)$$

So it must be  $\vec{c} = \vec{E}_{Re}$  and also  $\vec{c}' = \vec{E}_{Im}$

#### 2.1.1 Types of polarization

- Elliptical (different max values through all axes, to define it we need  $x, y$  axis +  $\delta$  tilt)
  - Right-handed (anti-clockwise)
  - Left-handed (clockwise)
- Circular (equal max values through all axes)
  - Right-handed (anti-clockwise)
  - Left-handed (clockwise)
- Linear, values space along a segment

### 2.2 Stokes vector

Given  $\vec{E} = \begin{bmatrix} E_1 \\ E_2 \end{bmatrix} = \begin{bmatrix} E_{Re} \\ E_{Im} \end{bmatrix}$ , we can define also the Stokes vector as:

$$\vec{S} \in \mathbb{R}^4 \Rightarrow \vec{S} = \begin{bmatrix} S_0 \\ S_1 \\ S_2 \\ S_3 \end{bmatrix} = \begin{bmatrix} |E_1|^2 + |E_2|^2 \\ |E_1|^2 - |E_2|^2 \\ 2\Re[E_1 E_2^*] \\ 2\Im[E_1 E_2^*] \end{bmatrix}$$

It can be defined also with a normalized version on the first term:

$$\vec{U}_s = \frac{1}{S_0} \cdot \begin{bmatrix} S_1 \\ S_2 \\ S_3 \end{bmatrix}$$

Assuming  $\omega$  is **constant** (monochromatic), we can also state:

$$S_0^2 = S_1^2 + S_2^2 + S_3^2$$

Given  $a = |E_1|, b = |E_2|$  we can write the unit-Stokes vector also as:

$$\vec{U}_s = \frac{1}{a^2 + b^2} \begin{bmatrix} (a^2 - b^2) \cos 2\delta \\ (a^2 - b^2) \sin 2\delta \\ 2ab \end{bmatrix}$$

Introducing ellipticity  $\epsilon : \tan \epsilon = \frac{b}{a}$  we can say:

$$\vec{U}_s = \begin{bmatrix} \cos 2\epsilon \cos 2\delta \\ \cos 2\epsilon \sin 2\delta \\ \sin 2\epsilon \end{bmatrix}$$

### 2.2.1 Intuition behind

To receive light, a photo diode measures  $|\vec{E}_{TOT}|^2 = \dots = |\vec{E}_1|^2 + |\vec{E}_2|^2 + 2\Re[\vec{E}_1 \cdot \vec{E}_2^*]$ .

If we have orthogonality  $[\vec{E}_1 \cdot \vec{E}_2^*] = 0$

Knowing that  $\vec{E}$  can be written as:

$$\vec{E} = \begin{bmatrix} A_1 e^{j\phi_1} \\ A_2 e^{j\phi_2} \end{bmatrix} \Rightarrow \vec{S} = \begin{bmatrix} A_1^2 + A_2^2 \\ A_1^2 - A_2^2 \\ 2A_1 A_2 \cos(\phi_1 - \phi_2) \\ 2A_1 A_2 \sin(\phi_1 - \phi_2) \end{bmatrix}$$

So that  $\vec{S}$  depends only on  $A_1, A_2$  and a phase difference  $\Delta\phi$

Working with light intensity and polarizers, we can also write:

$$\vec{S} = \begin{bmatrix} I_{0^\circ} + I_{90^\circ} \\ I_{0^\circ} - I_{90^\circ} \\ I_{45^\circ} - I_{135^\circ} \\ I_{RH} + I_{LH} \end{bmatrix}$$

Here  $S_0^2 = S_1^2 + S_2^2 + S_3^2$  doesn't work anymore  $\Rightarrow$  cause we don't have a monochromatic light.

### 2.2.2 Degree of Polarization (DOP) of light

$$DOP = \frac{\sqrt{S_1^2 + S_2^2 + S_3^2}}{S_0}$$

## 2.3 Poincaré sphere

It is a sphere where all the points identify a different polarization. We use the unit-Stokes vector.  $2\delta$  is the azimuth angle, and  $2\epsilon$  is the elevation angle on the sphere, while the radius is 1.

### 2.3.1 Special Cases

- $\epsilon = 0 \rightarrow$  any vector is on the equator
- $\epsilon = 0 \Rightarrow b = 0 \rightarrow$  ellipse collapses in a segment
  - if also  $a \rightarrow 0 \Rightarrow \epsilon \rightarrow \pm \frac{\pi}{2}$
- Emispheres:
  - Upper emisphere: all right-handed elliptical polarizations
  - Lower emisphere: all left-handed elliptical polarizations
- Poles
  - North-pole:  $\epsilon = \frac{\pi}{4}$  Right-handed circular polarization
  - South-pole:  $\epsilon = -\frac{\pi}{4}$  Left-handed circular polarization

### 3 Lecture - Oct. 16

#### 3.1 Total Internal Reflection

Between two different medium/materials:  $n_i \sin \theta_i = n_t \sin \theta_t$

In the special case where  $n_t < n_i$  we find **critical angle**:  $\theta_c = \arcsin \frac{n_t}{n_i}$

#### 3.2 Numerical Aperture

Suppose we have a fiber with core having index  $n_1$  and cladding  $n_2$ ,  $\beta_m$  is the entrance angle from air  $n_0$ : how small must be to trap light into the core? We know  $n_0 \sin \beta_m = n_1 \sin \beta_l$ , we want to satisfy the condition:  $\frac{\pi}{2} - \beta_l \geq \theta_c = \arcsin \frac{n_2}{n_1} \Rightarrow \sin(\frac{\pi}{2} - \theta_c) = \cos(\beta_l) \geq \frac{n_2}{n_1}$

$$NA = \sqrt{n_1^2 - n_2^2} \quad \text{such that} \quad \sin \beta_m \leq NA$$

For TLC purposes, better if  $NA \rightarrow 0$ , else there can be ISI caused by light going through different paths with same  $v_{\text{propagation}} = \frac{c_0}{n_1}$ . So better to keep  $n_1 - n_2$  small.

Considering Difference with straight path:

$$\Delta\tau = \frac{\Delta L}{v} = L \left( \frac{1}{\cos \beta_l} - 1 \right) \frac{n_1}{c}$$

we can see there is a  $\Delta\tau_{\text{max}}$ :

$$\Delta\tau_{\text{max}} \approx L \frac{(NA)^2}{2n_2c} \quad : \quad \Delta\tau_{\text{max}} \propto L \cdot (NA)^2$$

Where does  $\Delta\tau_{\text{max}}$  matter? To consider when  $\Delta\tau_{\text{max}}$  is about the same order of  $T_{BIT}$

##### 3.2.1 Maximum Fiber Bandwidth

$$B_{\text{max}} \equiv \frac{1}{\Delta\tau_{\text{max}}}$$

##### 3.2.2 Modal Dispersion

$$B_{\text{max}} \propto \frac{1}{L \cdot (NA)^2}$$

There are 2 ways we can increment bandwidth:

- Short  $L$
- reduce Numerical Aperture  $NA$

### 3.3 Fiber types

#### 3.3.1 Step-Index fibers

Refracting Index Profile is rectangular over the core.

#### 3.3.2 Graded-Index fibers

Refracting Index Profile is going as an exponential from a 'triangular' profile  $g = 0$  to the 'step-index' profile  $g = \infty$  over the core.

The refracting index goes like:

$$n^2(r) = n_1^2 \left[ 1 - 2\Delta \left( \frac{r}{a} \right)^g \right] \quad \text{with } \Delta = \frac{(NA)^2}{2n_1^2}$$

For example with  $g = 2$ ,  $\Delta\tau_{\text{max}} \propto L \cdot (NA)^4$  (4-th power!). Best selection of  $g$  is around  $\approx 1.9$ , the only parameter out of control is the core size  $\propto \approx 10\mu m$



### 3.4 Modal Theory

Solving EM problem for optical fibers is a linear combination of special solutions: the modes. Usually there are mainly 2 categories of modes:

- Guided Modes:  $\exists$  in a finite number
- Radiation Modes: not-finite number, not discrete, to neglect if we use right hypothesis (no fiber bending).

#### 3.4.1 Step-Index fibers' case

Assuming only Linearly polarized modes, we'll use:

$$TE_{o,p} \quad TM_{o,p} \quad HE_{m,p} \quad EH_{m,p}$$

Assuming:

$$\begin{cases} \vec{E}(x, y, z) = \vec{\mathcal{E}}(x, y)e^{-j\beta z} \\ \vec{H}(x, y, z) = \vec{\mathcal{H}}(x, y)e^{-j\beta z} \end{cases}$$

Recalling  $k_i^2 = \omega^2 \mu \epsilon = \omega^2 \frac{n_i}{c_0} = \frac{2\pi}{\lambda_0} n_i$  for each  $i$ -th region of the fiber, we get this modal set of equations:

- $j\beta E_z = \vec{\nabla} \cdot \vec{E}_t$
- $j\beta H_z = \vec{\nabla} \cdot \vec{H}_t$
- $j\omega\mu_0 \vec{H}_t = \vec{\mathcal{U}}_z \times (\nabla E_z + j\beta \vec{E}_t)$
- $\nabla^2 \vec{E}_t - (\beta^2 - k^2) \vec{E}_t = 0$

It can be proved that:

$$k_{\text{cladding}} < \beta < k_{\text{core}} \quad \Longleftrightarrow \quad n_{\text{cladding}} < n_{\text{effective}} < n_{\text{core}}$$

because of the  $k_i$  definition, and given that  $\beta = \frac{2\pi}{\lambda_0} n_{\text{eff}}$

Introducing other eigenvalues:

$$\chi_{\text{core}}^2 = k_{\text{core}}^2 - \beta^2 \quad \chi_{\text{cladding}}^2 = k_{\text{cladding}}^2 - \beta^2$$

we can also say that  $\chi_{\text{core}}^2 \geq 0, \chi_{\text{cladding}}^2 \leq 0 \Rightarrow \chi_{\text{cladding}} \in \mathbb{C}$

Now we have to solve the Helmholtz equation:

$$\nabla^2 \vec{E}_t + \chi^2 \vec{E}_t = 0 \quad \vec{E}_t = \begin{bmatrix} E_x \\ E_y \\ 0 \end{bmatrix}$$

Here comes the LP assumption  $\Rightarrow$  polarization stays the same everywhere on the cross section, hence:

$$\text{Y-polarization} \quad \Rightarrow \quad E_x = 0 \quad \Rightarrow \quad \vec{E}_t = \begin{bmatrix} 0 \\ E_y \\ 0 \end{bmatrix}$$

## 4 Lecture - Oct. 17

### 4.1 LP Approximation

At the boundary we need:

$$\text{Satisfaction of } \begin{cases} \vec{E}_{co,tan} = \vec{E}_{cl,tan} \Rightarrow \vec{E}_{co,y} = \vec{E}_{cl,y} \\ \vec{D}_{co,norm} = \vec{D}_{cl,norm} \Rightarrow \epsilon_{co}\vec{E}_{co,norm} = \epsilon_{cl}\vec{E}_{cl,norm} \end{cases}$$

In the end we can summarize this as  $\frac{(NA)^2}{n_{cl}^2} \ll 1$ , for this reason we use **Weakly guiding fibers**.

In the fiber cross-section domain  $\{r, \phi, z\}$  we have:

$$\nabla^2 = \frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \phi^2} + \frac{\partial^2}{\partial z^2}$$

Applying the laplacian equation to  $E_y$ :  $\nabla^2 \vec{E}_y + \chi^2 \vec{E}_y = 0$  we get:

$$\nu^2 = -\frac{1}{g} \frac{d^2 g}{d\phi^2} = \frac{r^2}{f} d^2 f dr^2 + \frac{r}{f} df dr + r^2 \chi^2$$

After the resolution of the obtained Harmonic Oscillator diff. eq., and assuming  $\nu = n \in \mathbb{N}$  we get also:

$$g_{co}(\phi) = C_{co} \cos(n\phi + \phi_{0,co}) \quad g_{cl}(\phi) = C_{cl} \cos(m\phi + \phi_{0,cl})$$

#### 4.1.1 LP Approximation using Bessel functions

- **Core**

$$f(r) = q_1 J_n(\chi_{co} r) \Rightarrow E_{y,co} = q_{co} J_n(\chi_{co} r) \cos(n\phi + \phi_{co})$$

- **Cladding**

$$\begin{cases} \tilde{\chi}_{cl} = \sqrt{-\chi_{cl}^2} \\ f(r) = b_1 K_m(\tilde{\chi}_{cl} r) \end{cases} \Rightarrow E_{y,cl} = b_1 K_m(\tilde{\chi}_{cl} r) \cos(m\phi + \phi_{cl})$$

We need that for each value  $m = n$  and  $\phi_{co} = \phi_{cl} = \phi_0$ , so:

$$\frac{q_{co}}{q_{cl}} = \frac{\epsilon_{cl} K_n(\tilde{\chi}_{cl} a)}{\epsilon_{co} J_n(\chi_{co} a)} \Rightarrow \vec{E}_t(r, \phi) = \begin{cases} q \frac{J_n(\chi_{co} r)}{J_n(\chi_{co} a)} \cos(n\phi + \phi_0) & \text{if core } (r < a) \\ q \frac{K_n(\tilde{\chi}_{cl} r)}{K_n(\tilde{\chi}_{cl} a)} \cos(n\phi + \phi_0) & \text{if cladding } (r \geq a) \end{cases}$$

#### 4.1.2 LP Mode Characteristic Equation

$$LP_{n,p} \quad \Longleftrightarrow \quad \chi_{co} a \frac{J_{n+1}(\chi_{co} a)}{J_n(\chi_{co} a)} = \tilde{\chi}_{cl} a \frac{K_{n+1}(\tilde{\chi}_{cl} a)}{K_n(\tilde{\chi}_{cl} a)}$$

How many solutions should I count?

Some don't have physical meaning, so we impose  $k_{cl} < \beta < k_{co} \Rightarrow n_{cl} < n_{eff} < n_{co}$

As  $n_{co} \rightarrow n_{cl}$  light tries to escape into the cladding, thus each mode has a Cut-off frequency.

### 4.2 Cut-Off Frequency

We impose:

$$\beta = k_{cl} \quad \Rightarrow \quad \begin{cases} \chi_{co} = \sqrt{k_{co}^2 - \beta^2} = \dots = \frac{2\pi}{\lambda_0} NA \\ \tilde{\chi}_{cl} \rightarrow 0 \end{cases}$$

Knowing the limit  $\lim_{u \rightarrow 0} \frac{u k_{n+1}(u)}{k_n(u)} = 2n$ :

$$\omega \text{ cut-off frequency} \quad : \quad \frac{\omega a}{c_0} (NA) \frac{J_{n+1}\left(\frac{\omega a}{c_0} (NA)\right)}{J_n\left(\frac{\omega a}{c_0} (NA)\right)} = 2n$$

Which are the first solutions?

$$LP_{0,1} = \begin{cases} n = 0 \\ \omega_c = 0 \end{cases} \quad LP_{1,1} = \begin{cases} n = 1 \\ \omega_{1,1} \cong \frac{2.405 c_0}{a(NA)} \end{cases}$$

With Cut-off frequency we block propagation of other modes except  $LP_{0,1}$ .

## 5 Lecture - Oct. 23

### 5.1 Modal Dispersion Diagram

We have to introduce two normalized variables:

- **Normalized Frequency  $\nu$**

$$\nu = a\sqrt{\chi_{co}^2 + \tilde{\chi}_{cl}^2} = 2\pi\frac{a}{\lambda}NA$$

- **Normalized propagation constant  $b$**

$$b = \frac{\beta^2 - k_{cl}^2}{k_{co}^2 - k_{cl}^2} = \frac{n_{eff}^2 - n_{cl}^2}{(NA)^2}$$

So that we obtain two other equivalences:

$$a\chi_{co} = \nu\sqrt{1-b} \qquad a\tilde{\chi}_{cl} = \nu\sqrt{b}$$

Having  $\nu$  on the  $x$ -axis, and  $b$  on the  $y$ -axis, we find the Dispersion Diagram: we see the cutoff  $\omega$  as intersections with  $b = 0$ , and for a given  $\lambda$  window we can see also the available modes propagating into the fiber.

#### 5.1.1 Amount of power by each mode

We know the Poynting vector is  $\vec{P} = \dots = \frac{1}{2} \frac{\beta}{\omega\mu_0} |\vec{E}_t|^2 \vec{U}_z$ , thus  $\vec{P} \propto |\vec{E}|^2$ .  
Due to orthogonality we have this implication:

$$\vec{E}_{tot} = \sum_n c_n \vec{E}_n e^{-j\beta_n z} \quad \Rightarrow \quad |\vec{E}_{tot}|^2 = \sum_n |c_n|^2 |\vec{E}_n|^2$$

### 5.2 Losses on Fiber

#### 5.2.1 Attempt: Launching a beam over a fiber

We have to decide the right  $c_n$  to check which modes are excited.  
By projecting total field on different nodes we get the  $c_n$ :

$$\text{Superposition Integral} \rightarrow \vec{E}_{tot} = \sum_n c_n \int (\vec{E}_n \times \vec{H}_n^*) \vec{U}_z dS$$

Why light keep bouncing on and on? Considering only 2 modes:

$$E_{tot} = c_{0,1} \vec{E}_{0,1} e^{-j\beta_{0,1} z} + c_{1,1} \vec{E}_{1,1} e^{-j\beta_{1,1} z} = e^{-j\beta_{0,1} z} \left\{ c_{0,1} \vec{E}_{0,1} + c_{1,1} \vec{E}_{1,1} e^{-j(\beta_{1,1} - \beta_{0,1})z} \right\}$$

We can see the period of bouncing is given by  $\frac{2\pi}{\beta_{1,1} - \beta_{0,1}} = \frac{\lambda}{n_{1,1} - n_{0,1}}$ , this gives ISI.

#### 5.2.2 Attenuation

For little intervals of  $z$ , we can Taylor-expand the power as:

$$P(z + \Delta z) \cong P(z) + \frac{dP}{dz} \Delta z = (1 - \alpha(z)\Delta z)P(z)$$

Thus we conclude that, in a fiber, power decreases **exponentially**, so:

$$P(z) = P(0) \exp \left\{ - \int_0^z \alpha(z') dz' \right\} \quad \Rightarrow \quad P_{dBm} = 10 \log_{10} \frac{P(z)}{1 \text{ dBm}} = P_{dBm}(0) - \alpha z \cdot 10 \log_{10} e$$

Main causes of attenuation are:

- **Electronic Transitions** (mostly in UV bandwidth, limit BW in higher  $\lambda$ s)
- **Molecular Vibrations** due to non-rigid structure of  $SiO_2$ , defines 3rd/4th main TX windows
- **Rayleigh Scattering** for little  $\lambda$ s, when particles sizes  $\ll \lambda$ ,  $P_{scattering} \propto \frac{1}{\lambda^4}$

## 6 Lecture - Oct. 24

### 6.1 Losses on Fiber

#### 6.1.1 Absorption

Main causes of power absorption are:

- $OH^-$ , peak is between 2nd and 3rd window
- $H_2$ , due to plastic degradation or production processes

#### 6.1.2 Bending Loss

Since we know propagations on a fiber depends on  $e^{-j\beta z}$ , which is associated to  $v_p = \frac{\omega}{\beta}$  for  $k_{cl} < \beta < k_{co}$  we can also write, using  $R$  as curvature radius of fiber bending:

$$v_p(x) = v_p(0) \frac{R+x}{R}$$

Since we want  $\beta > k_{cl}$  such that  $\exists x_c : v_p(x_c) = \frac{\omega}{k_{cl}}$ , we can define:

$$x_c = R \left( \frac{\beta - k_{cl}}{k_{cl}} \right) \quad \text{NOTE : } x_c \text{ depends on } \lambda$$

NOTE: if  $x_c \gg 0$  (is outside of fiber core radius) we can bend the fiber without noticeable losses. We can finally quantify bending loss as:

$$\alpha_{bend} = C e^{-2\sigma x_c}$$

with  $2\sigma$  the decendency of Bessel Function after core

#### 6.1.3 Cutoff Frequency Identification Standard - Idea

$\lambda$  close to cutoff condition  $\leftrightarrow$  more subject to bending.

We send a broadband light source signal through a fiber towards an Optical Signal analyser, we measure one time with no bending, then we add a bent in the fiber path. Making the difference between these 2 graphs we find another graph where, taking the 2nd  $y = 0.1dB$  intersection with the obtained function, we get the cutoff  $\lambda$  for many modes.

#### 6.1.4 Microbending

It happens when a fiber is wavy, not so straight. At some point we can have disruption of service.

## 6.2 Transmitting a signal on a fiber

### 6.2.1 Frequency Domain

We have a signal:

$$\vec{E}(x, y, z, \omega) = A(\omega) \sum_{n=1}^M c_n \vec{E}_n(x, y) \exp \{-jz\beta_n(\omega)\}$$

Since we want to work with the baseband signal, we have to work in  $\omega + \omega_0$ :

$$\vec{E}(x, y, z, \omega) = A(\omega) \sum_{n=1}^M c_n \vec{E}_n(x, y) \exp \{-jz\beta_n(\omega + \omega_0)\}$$

### 6.2.2 Time domain

How do we get back to time domain? We have to Taylor-expand  $\beta$  around  $\omega_0$ :

$$\beta_n(\omega + \omega_0) \approx \beta_n(\omega_0) + \left. \frac{d\beta_n}{d\omega} \right|_{\omega_0} \omega = \beta_{n,0} + \beta_{n,1}\omega$$

Defining **Group Velocity** as speed at which information is transmitted:  $v_f = 1/\frac{d\beta}{d\omega}$ , and **Group Delay**  $\tau_g = z\beta_{n,1}$  as delay signal accumulates during propagation, we can state:

$$e(t) = \sum_n c_n E_n(x, y) a(t - \tau_g) e^{-j\beta_{n,0}z} \quad \text{no more } \omega \text{ dependency}$$

## 7 Lecture 30. Oct

### 7.1 Chromatic Dispersion

Depends on 3 factors: material dispersion (positive) + guide dispersion (negative) + profile dispersion (negligible)

#### 7.1.1 Definition

$$\beta_2 = \frac{d^2\beta}{d\omega^2} = \frac{d}{d\omega} \left( \frac{1}{v_g} \right) \quad \left[ \frac{ps^2}{km} \right] \equiv \left[ \frac{\frac{ps}{THz}}{km} \right]$$

$$D = \frac{d}{d\lambda} \left( \frac{d\beta}{d\omega} \right) = -\frac{2\pi c_0}{\lambda^2} \beta_2 \quad \left[ \frac{ps}{nm \cdot km} \right]$$

#### 7.1.2 Time Domain

Assuming we use **Gaussian pulse!!**

$$\bar{e}(t) = \frac{A_0}{\sqrt{1 + j\frac{L\beta_2}{2\sigma^2}}} \bar{E} \exp \{ -j\beta_0 L \} \exp \left\{ j \frac{(t - L\beta_1)^2}{2L\beta_2 \left( 1 + \frac{4\sigma^2}{L^2\beta_2^2} \right)} \right\} \exp \left\{ -\frac{(t - L\beta_1)^2}{4\sigma^2 \left( 1 + \frac{L^2\beta_2^2}{4\sigma^4} \right)} \right\}$$

Intuition: all goes as  $\frac{L^2\beta_2^2}{4\sigma^4}$ , if numerator increase or denominator decreases = amplitude decreases. (Area under pulse remains constant, since no losses)

#### 7.1.3 Compensation Methods

If I want to transmit a wide spectrum, we know that if we use smaller portions of the spectrum  $v_g$  remains the same, but if we use a wide spectrum  $v_g(\omega_i) \neq v_g(\omega_j)$ , so we must compensate these group velocities.

Solution: use a second fiber with different characteristics. Signal after the 2 fibers has this form:

$$E(\omega) = E_0(\omega) e^{-j\beta_A(\omega)L_A} e^{-j\beta_B(\omega)L_B}$$

expanding phase term we see that, for second order (Chromatic Dispersion), from Taylor expansion we need:

$$\beta_{A,2}L_A + \beta_{B,2}L_B = 0 \quad \implies \quad L_B = -\frac{\beta_{A,2}}{\beta_{B,2}}L_A$$

Typically there exists DCF<sup>1</sup> that have a  $\beta_2$  compatible with this condition.

Gaussian pulse input signals after the fiber get up/down 'chirped' (frequency start low and then rises/decreases quickly)

#### 7.1.4 Nobel Idea

If I have short pulse and low energy how can I amplify it?

Processing Chain:

- Input: Low energy pulse
- Stretcher: fiber with chromatic dispersion (increment bandwidth)
- Wideband Amplifier
- Compressor: DCF fiber to compensate stretcher effect
- Output: High energy pulse

#### 7.1.5 Slope of Chromatic Dispersion

Trying to increase BW, third-order effect in Taylor expansion of previous step. We need:

$$\frac{L_B}{L_A} = -\frac{\beta_{A,2}}{\beta_{B,2}} = -\frac{\beta_{A,3}}{\beta_{B,3}}$$

I telecommunication miracle: total dispersion summing up all is 0 near the 2TX Window.

<sup>1</sup>Dispersion Compensating Fibers

## 8 Lecture 31. Oct

### 8.1 Chromatic Dispersion

#### 8.1.1 Measurement strategy

Knowing that

$$DL = \frac{d}{d\lambda} \left( \frac{L}{v_g} \right) = \frac{d\tau_d}{d\lambda}$$

How do we measure chromatic dispersion? We get the delay between the reference signal and the received signal at photodiode, and then we plot group delay  $\tau_d$  as a function of  $\lambda$ , the  $\lambda_0$  where the DL as derivative of  $\tau_d$  over  $\lambda$  crosses the x-axis is called **Zero Dispersion Wavelength**. The slope of the DL goes as  $\frac{1}{6}\omega^3 \frac{d^3\beta}{d\omega^3} \big|_{\omega_0}$

### 8.2 Polarization Mode Dispersion - PMD

Origins: mechanical stresses of the fiber, external magnetic fields. It starts when the circular symmetry is broken. There is no way to compensate it.

#### 8.2.1 Jones Matrix

We know that, in **Constant Asymmetry** conditions:

$$\vec{E}(r, \phi, z) = (a_x(z)\vec{\mathcal{U}}_x + a_y(z)\vec{\mathcal{U}}_y)f(r) \quad \text{with} \quad \begin{cases} a_x(z) = a_x(0)e^{-j\beta_x z} \\ a_y(z) = a_y(0)e^{-j\beta_y z} \end{cases}$$

We can define Jones matrix as:

$$J(z) = \begin{bmatrix} e^{-j\beta_x z} & 0 \\ 0 & e^{-j\beta_y z} \end{bmatrix}$$

such that

$$\vec{a}(z) = \begin{bmatrix} a_x(z) \\ a_y(z) \end{bmatrix} = J(z)\vec{a}(0)$$

#### 8.2.2 Euler Matrix

Recalling Stokes vector  $\vec{s}$ , the analogous matrix in this case is the Euler matrix defined such that:

$$\hat{s}(z) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \Delta\beta z & \sin \Delta\beta z \\ 0 & -\sin \Delta\beta z & \cos \Delta\beta z \end{bmatrix} \hat{s}(0)$$

The state of polarization is represented as concentric circles around the main axis. So polarization draws circles.

#### 8.2.3 Frequency Dependency

Adding frequency dependency:

$$\vec{a}(z, \omega) = J(z, \omega)\vec{a}(0, \omega)$$

There is a change in  $\beta_x$ , which need to be expanded as per Taylor-expansion method.

Since  $\beta_{x,1} = \frac{1}{v_{x,g}}$ , we have  $\begin{cases} \beta_x(\omega) \approx \beta_{x,0} + \omega\beta_{x,1} \\ \beta_y(\omega) \approx \beta_{y,0} + \omega\beta_{y,1} \end{cases}$

So we have to use a Jonas-like matrix for every order of Taylor-expansion:

$$\hat{s}(L, \omega) = R_0(L) \cdot R_1(L, \omega) \cdot \hat{s}(0, \omega)$$

where  $R_0$  and  $R_1$  are defined as:

$$R_0(L) = \begin{bmatrix} e^{-j\beta_{x,0}L} & 0 \\ 0 & e^{-j\beta_{y,0}L} \end{bmatrix} \quad R_1(L, \omega) = \begin{bmatrix} e^{-j\omega\beta_{y,1}L} & 0 \\ 0 & e^{-j\omega\beta_{y,1}L} \end{bmatrix}$$

Given  $G(\omega)$  the spectrum of the signal on the fiber, we also know that  $\vec{a}_{out}(\omega) = \mathbf{U}(\omega)\vec{a}_{in}(0)G(\omega)$ , so we have to find the equivalent matrix  $\mathbf{U}$  such that:

$$\vec{a}(L, \omega) = \mathbf{U}(L, \omega)\vec{a}(0, \omega)$$

### 8.2.4 Time Domain

We know  $\mathbf{U}$  is a unitary matrix ( $\mathbf{U}^{-1} = \mathbf{U}^*$ )

Deriving on frequency we obtain:

$$\frac{dU}{d\omega} = -jQ(\omega)\mathbf{U}(\omega) \quad \Rightarrow \quad Q(\omega) = j\frac{dU}{d\omega}\mathbf{U}^{-1}$$

using the analogy with:

$$\frac{d}{d\omega}e^{-j\beta(\omega)L} = -j\left(\frac{d\beta}{d\omega}L\right)e^{-j\beta(\omega)L} \quad \Rightarrow \quad \begin{cases} Q \equiv \frac{d\beta}{d\omega}L \\ \mathbf{U} \equiv e^{-j\beta(\omega)L} \end{cases}$$

Due to polarization effect, we only expand with Taylor to the first term  $Q(\omega) \approx Q_0$ , so:

$$\frac{dU}{d\omega} = -jQ_0\mathbf{U}(\omega) \quad \Rightarrow \quad \mathbf{U}(\omega) = \mathbf{U}(\omega_0)e^{-jQ_0\omega}$$

Using  $Q$  hermitianity:

$$Q\hat{v}_k = \tau_k\hat{v}_k, \tau_k \in \mathbb{R}$$

## 9 Lecture 6. Nov

### 9.1 Polarization Mode Dispersion - PMD

#### 9.1.1 Principal States of Polarization

From the previous lecture, and the **analogy!!**:

$$\underline{\mathbf{Q}}_0 \hat{v}_k = \tau_k \hat{v}_k \quad \Longleftrightarrow \quad e^{-j\underline{\mathbf{Q}}_0 \omega} \hat{v}_k = e^{-j\tau_k \omega} \hat{v}_k$$

$\hat{a}_{out}$  could be written as:

$$\hat{a}_{out}(\omega) \cong G(\omega) e^{-j\underline{\mathbf{Q}}_0 \omega} (\underline{\mathbf{U}}_0 \hat{a}_{in})$$

where the last term ( $\underline{\mathbf{U}}_0 \hat{a}_{in}$ ) is the output polarization at carrier frequency, we can write the  $\hat{a}_{out}$  as:

$$\hat{a}_{out}(t) = c_1 \hat{v}_1 g(t - \tau_1) + c_2 \hat{v}_2 g(t - \tau_2)$$

Note that  $\hat{v}_1, \hat{v}_2$  are the **Principal States of Polarization - PSP**.

There is this implication:

$$\text{NO losses} \Rightarrow \underline{\mathbf{U}} \text{ unitary} \Rightarrow \underline{\mathbf{Q}} \text{ hermitian} \Rightarrow \begin{cases} \tau_k \text{ real} \\ \hat{v}_1^* \cdot \hat{v}_2 = 0 \end{cases} \quad \text{orthogonality}$$

PSP with Differential Group Delay (DGD)  $\Delta\tau = |\tau_1 - \tau_2|$  are considered/represented by the PMD matrix.

#### 9.1.2 PMD vector

What if I have a fiber and i add a piece  $\Delta L$ , how does PMD change?  $Q_0$  has an implicit  $L$ : problem.

Looking to the Stokes analogy:  $\vec{a}_{out} = \underline{\mathbf{U}}(\omega) \vec{a}_{in} \Leftrightarrow \hat{s}_{out} = \underline{\mathbf{R}}^{3 \times 3}(\omega) \hat{s}_{in}$ , we can define a vector:

$$\exists \text{ PMD Vector } \vec{\Omega} \quad : \quad \frac{d\hat{s}_{out}}{d\omega} = \vec{\Omega} \times \hat{s}_{out}$$

Recap of the analogy:

$$\underline{\mathbf{Q}}(\omega) \rightarrow \begin{cases} \tau_k \leftrightarrow \text{DGD} \\ \hat{v}_k \leftrightarrow \text{PSP} \end{cases} \quad \Leftrightarrow \quad \vec{\Omega}(\omega) \rightarrow \begin{cases} |\vec{\Omega}| = \Delta\tau \leftrightarrow \text{DGD} \\ \hat{\Omega} = \frac{\vec{\Omega}}{|\vec{\Omega}|} \leftrightarrow \text{PSP} \end{cases}$$

We can also say that, for a  $n$ -cut parts fiber:

$$\vec{\Omega}_n = \vec{\Lambda}_n + \underline{\mathbf{R}}_n \vec{\Omega}_{n-1} \quad \text{with} \quad \begin{cases} \vec{\Lambda}_n \rightarrow \text{PMD vector of only last } n\text{-th sector} \\ \vec{\Omega}_{n-1} \rightarrow \text{total PMD vector of the prev. sectors} \end{cases}$$

#### 9.1.3 Estimation of total DGD value

Looking at the average of DGD squared:

$$\langle \Delta\tau^2 \rangle = \langle DGD^2 \rangle = \langle \delta\tau_n^2 \rangle = \langle |\Omega_n|^2 \rangle = \langle |\vec{\Lambda}_n|^2 \rangle + \langle |\vec{\Omega}_{n-1}|^2 \rangle + 2 \langle \vec{\Lambda}_n \underline{\mathbf{R}}_n \vec{\Omega}_{n-1} \rangle \rightarrow 0$$

So for a  $n$ -segment fiber, we have that there is a linear dependency on the single segment  $\delta\tau_k$ :

$$\langle \Delta\tau_n^2 \rangle = \sum_{k=1}^n \langle \delta\tau_k^2 \rangle \quad \Rightarrow \quad \langle \Delta\tau_n^2 \rangle = n \langle \delta\tau^2 \rangle$$

Going back to DGD value, using square root we have a  $\langle \Delta\tau_n \rangle \propto \sqrt{n}$  dependency.



## 10 Lecture 7. Nov

### 10.1 Optical Amplifiers

There can be electrical-based amplifiers, such that they amplify only electrically channels of  $< 100\text{GHz}$  BW by WDM-DeMuxing and ReMuxing them, but the best solution is to use directly Optical ones (at light level).

#### 10.1.1 Classes

From the least popular to the most popular one:

- Parametric Amplifiers: least noisy ones, non-linear Kerr effect
- Scattering Amplifiers/Raman Amplifiers: based on Raman Scattering,  $\text{SiO}_2$  resonance
- Semiconductor Amplifiers: modulating another carrier based on our signal, used in FTTH, low cost
- **Inversion Population Amplifiers**: fibers with doped elements, for long paths (Erbium: C+L bands)

### 10.2 Light-Matter Interactions

Assume a photon has an energy  $E = h\nu$ , where  $h = 6.6 \cdot 10^{-34} [\text{J} \cdot \text{s}]$  is the Planck constant and  $\nu$  is the frequency  $\approx 193\text{THz}$  at  $\lambda = 1550\text{nm}$ . Atoms can have discrete values of levels of energy, so electrons can jump from one layer to another.

Given that we have 2 layers, #1 is the bottom one, #2 is the top one, we can see different phenomena, reported in the following subsections.

#### 10.2.1 Simulated Absorption

Since  $N_2$  is the number of atoms in a given volume, population at level 2 changes as:

$$\frac{dN_2}{dt} = \sigma_{1,2} \Phi N_1 \quad \text{with} \quad \begin{cases} \sigma_{1,2} = \text{cross section} \\ \Phi = \text{flux: } [(\# \text{photons}/s)/m^2] \\ N_1 = \text{population of bottom layer} \end{cases}$$

So the entering photons disappear passing energy to promote electrons from the bottom layer.

#### 10.2.2 Simulated Emission

$$\frac{dN_2}{dt} = -\sigma_{2,1} \Phi N_2 \quad \text{with} \quad \begin{cases} \sigma_{2,1} = \text{cross section} \\ \Phi = \text{flux: } [(\# \text{photons}/s)/m^2] \\ N_2 = \text{population of top layer} \end{cases}$$

Basically electrons decay generating duplicating the entering photons. In general  $\sigma_{1,2} \neq \sigma_{2,1}$

#### 10.2.3 Spontaneous Emission

Electrons from the top layer decay generating a photon of energy  $h\nu = E_2 - E_1$ .

It is the source of noise in amplification.

$$\frac{dN_2}{dt} = -\frac{1}{\tau_{sp}} N_2$$

$\tau_{sp}$  is the energy-level mean lifetime, average time an electron can stay in that level before decaying. Photons emitted have random properties, unknown frequency, power, direction etc.

#### 10.2.4 3-Level Systems

Assume we have 3 levels, with many electrons that can move between the levels (given by doping material).

- From L3 (top)  $\rightarrow$  L2 (middle): L3 instable, we have a quick decay to L2
- L2: Metastable, we need a sufficiently large  $\tau_{sp}$  to store the electrons there
- Pump-photon:  $h\nu = E_3 - E_1$ , they promote electrons from L1 to L3

If we pump fast enough we make L2 full of electrons, and we reach a condition of **Population Inversion**, so there is amplification by stimulated emissions.

II Telecommunication Miracle: Erbium has  $\lambda_{pump} = 1480, 980\text{nm} \approx 1550$ , near our TX window.

## 10.3 Erbium-doped fiber amplifiers

### 10.3.1 Fundamental Components

- of course the Erbium Doped Fiber (EDF)
- Pump laser, source of energy for amplification
- Dichroic Coupler, like a mixer of signals
- 2 isolators, to avoid feedback loops (1 before and 1 after amplification stadium)

### 10.3.2 Rate Equations

What does it mean ‘to pump fast enough’? Suppose we have an infinitesimal cylindre of fiber:

- surface  $s$
- length  $\Delta z$
- in-flux  $\Phi(z, t)$
- out-flux  $\Phi(z + \Delta z, t + \Delta t (= \frac{\Delta z}{c}))$

So a first rate equation is given by **known in-flux + contribute of stimulated emissions - contribute of stimulated absorption - expected out-flux = 0**

$$s\Delta t \left( \phi_s(z, t) - \phi_s \left( z + \Delta z, t + \frac{\Delta z}{c} \right) \right) + (s\Delta z) \Delta t [\sigma_s \phi_s(z, t) (N_2(z, t) - N_1(z, t))] = 0$$

At the end, after Taylor expansions, and knowing that  $I_{\square} = h\nu_{\square}\phi_{\square}$ :

$$\text{Pump : } \frac{\partial I_p}{\partial z} + \frac{1}{c} \frac{\partial I_p}{\partial t} = -\sigma N_1 I_p$$

$$\frac{\partial I_s}{\partial z} + \frac{1}{c} \frac{\partial I_s}{\partial t} = \sigma(N_2 - N_1)I_s$$

**Balance of population density**, we need to consider also spontaneous emissions. Since  $N_{tot} = N_1 + N_2$ :

$$\frac{\partial N_2}{\partial t} = \frac{\sigma_p I_p}{h\nu_p} N_1 - \frac{\sigma_s I_s}{h\nu_s} (N_2 - N_1) - \frac{1}{\tau_{sp}} N_p$$

## 11 Lecture 13. Nov

### 11.1 Rate Equations for 3-Level system

#### 11.1.1 Population Inversion Condition

After defining saturation intensity for signal and pump as:

$$I_{S,SAT} = \frac{h\nu_s}{\sigma_s \tau_{sp}} \quad I_{P,SAT} = \frac{h\nu_p}{\sigma_p \tau_{sp}}$$

From previous lecture we can assume:

$$N_1 = N_{TOT} \frac{1 + \frac{I_S}{I_{S,SAT}}}{1 + \frac{I_P}{I_{P,SAT}} + 2 \frac{I_S}{I_{S,SAT}}} \quad N_2 = N_{TOT} \frac{\frac{I_P}{I_{P,SAT}} + \frac{I_S}{I_{S,SAT}}}{1 + \frac{I_P}{I_{P,SAT}} + 2 \frac{I_S}{I_{S,SAT}}}$$

We have to reach the population inversion condition (equivalence derived from next point):

$$N_2 > N_1 \quad \Longleftrightarrow \quad I_P > I_{P,SAT}$$

Trick: keeping same  $\nu_p$  if we increase  $\sigma_p$  we need less photons to generate inversion.

#### 11.1.2 Stationary Regime

$$\begin{aligned} \frac{dI_P}{dz} &= -\sigma_p N_{TOT} I_P \cdot \frac{1 + \frac{I_S}{I_{S,SAT}}}{1 + \frac{I_P}{I_{P,SAT}} + 2 \frac{I_S}{I_{S,SAT}}} \quad (\text{exp. decay, derivative always } < 0) \\ \frac{dI_S}{dz} &= -\sigma_s N_{TOT} I_S \cdot \frac{1 - \frac{I_P}{I_{P,SAT}}}{1 + \frac{I_P}{I_{P,SAT}} + 2 \frac{I_S}{I_{S,SAT}}} \Rightarrow \left( \frac{dI_S}{dz} > 0 \Leftrightarrow I_P > I_{P,SAT} \right) \end{aligned}$$

There is always an optimal length ( $L_{OPT}$ ) where signal starts to decay after gain phase.

### 11.2 Erbium Properties

#### 11.2.1 Gain

$$G = \frac{P_{OUT}(\nu_s)}{P_{IN}(\nu_s)}$$

#### 11.2.2 Power Conversion Efficiency

$$PCE = \frac{P_{S,OUT}(\nu_s) - P_{S,IN}(\nu_s)}{P_{P,IN}(\nu_p)} = \frac{\nu_s}{\nu_p} \cdot \frac{\Phi_{S,OUT} - \Phi_{S,IN}}{\Phi_{P,IN}} \leq \frac{\nu_s}{\nu_p} = \frac{\lambda_p}{\lambda_s}$$

#### 11.2.3 Saturation Input(Output) Power

For a given gain we have a max input power, viceversa for output power is the power I can measure before saturation.

#### 11.2.4 EDFA Bandwidth

Shape due to  $\sigma_s$ , kind of a mountain shape on a wavelength $_{\mu m}$ -over-gain $_{dB}$  chart.

Main problem here is ripple gain in a chain of EDFA (max distance is 100km, cause max gain is around 20dB).

#### 11.2.5 Noise: Amplified Spontaneous Emissions (ASE)

Using  $\langle n_s \rangle$  average numbers of photons at frequency  $\nu_s$  (signal + ASE):

$$\frac{d\langle n_s \rangle}{dz} = \sigma_s (N_2 - N_1) \langle n_s \rangle + 2\sigma_s N_2$$

$2\sigma_s N_2$  is the noise term due to ASE (kind of photons generate per unit of length). Factor 2 is due to the number of modes propagating in the fiber. Thus, in this case:

$$\langle n_s(z) \rangle = G(z) \langle n_s(0) \rangle + 2\eta(z)$$

with the gain defined as:

$$G(z) = \exp \left\{ \sigma_s \int_0^z N_2(z') - N_1(z') dz' \right\}$$

Power of ASE is:

$$P_{\text{ASE}}(z) = 2\eta(z)h\nu_s\Delta\nu_s \quad \text{with} \quad \begin{cases} \Delta\nu_s \rightarrow \text{Bandwidth} \\ 2\eta \rightarrow \text{number of photons} \\ h\nu_s \rightarrow \text{energy/photon} \end{cases}$$

I consider only noise in around  $\nu_s$  because it is where I'm interested.

### 11.2.6 Optical SNR

$$\text{OSNR} = \frac{\text{Power of the optical signal}}{\text{Power of the optical noise}}$$

### 11.2.7 Noise Figure

$$\text{NF} = \frac{\text{OSNR}_{\text{INPUT}}}{\text{OSNR}_{\text{OUTPUT}}} > 1$$

### 11.2.8 Noise Figure in EDFA case

Since:

$$\text{OSNR}_{in} = \frac{P_{IN}}{h\nu_s\Delta\nu_s} \quad \text{OSNR}_{out} = \frac{P_{OUT}}{P_{ASE} + h\nu_s\Delta\nu_s}$$

we have that (!EXAM)

$$\text{NF} = \frac{P_{ASE} + h\nu_s\Delta\nu_s}{Gh\nu_s\Delta\nu_s} = \frac{2\eta + 1}{G} \quad \Rightarrow \quad 2\eta = (G \cdot \text{NF} - 1) \quad \Rightarrow \quad P_{ASE} = (G \cdot \text{NF} - 1)h\nu_s\Delta\nu_s$$

Assuming  $N_1, N_2$  do not depend on  $z$  (constant):

$$\Rightarrow \text{NF} \approx \frac{1}{G} \left( 1 + 2(G-1) \frac{N_2}{N_2 - N_1} \right) \quad \text{for} \quad \begin{cases} G \gg 1 \\ N_2 \gg N_1 \end{cases} \quad \Rightarrow \text{NF} \approx 2$$

### 11.2.9 Amplifier Classes

In general: higher gains  $\Leftrightarrow$  lower NF

- Boosters: not so high gains, high NF
- Line Amplifiers: high gains from little signals, lower NF

## 12 Lecture 14 Nov.

### 12.1 General Rules of Amplification

#### 12.1.1 I: Noise Figure and concatenation

$$NF = \frac{\text{OSNR}_{\text{in}}}{\text{OSNR}_{\text{out}}} = \frac{\left(\frac{P_{in}}{h\nu_s\Delta\nu}\right)}{\left(\frac{G_1 G_2 P_{in}}{P_{ASE,2} + G_2 P_{ASE,1} + h\nu_s\Delta\nu}\right)}$$

Noise figure of concatenated amplification blocks:

$$NF_{TOT} = NF_1 + \sum_{i=2}^m \frac{NF_i - 1}{\left(\prod_{j=1}^{i-1} G_j\right)}$$

#### 12.1.2 II: Amplifier Location

If we put the EDFA block right after the TX, we get:

$$NF_{TOT} = NF + \frac{1-A}{AG} \quad \Rightarrow @ \text{ RX: } P_{out} = (GP_{in} + P_{ASE})A \quad \Rightarrow \text{ASE power is attenuated by A}$$

If we put the EDFA block after the fiber path, before the RX:

$$NF_{TOT} = \frac{NF}{A} \quad \Rightarrow @ \text{ RX: } P_{out} = (GA \cdot P_{in}) + P_{ASE} \quad \Rightarrow \text{ASE power is NOT attenuated by A}$$

#### 12.1.3 Noise Figure Measurement

$$P_{ASE} = (G \cdot NF - 1)h\nu_s\Delta\nu \quad \Rightarrow \quad NF = \frac{\frac{P_{ASE}}{\Delta\nu} + h\nu_s}{h\nu_s} = 1 + \frac{P_{ASE}}{h\nu_s\Delta\nu}$$

We have to consider noise over a certain bandwidth.

## 13 Lecture 4 Dec.

### 13.1 Non Linear Effects

#### 13.1.1 Electron position modeling

For a Non-conductive media, mechanic oscillation equation (electron can be modeled as a mass + spring):

$$m\vec{a} = -e\vec{E} - e\vec{v} \times \vec{B} - k\vec{r} - m\Gamma\vec{v} \quad \rightarrow \quad \text{Coulomb, Lorentz, Elastic and Friction contribution}$$

After some simplifications we get:

$$m \left[ \frac{d^2x}{dt^2} + 2\Gamma \frac{dx}{dt} + \Omega^2 x \right] = -e\vec{E}(t) \quad \text{with} \quad \begin{cases} \Gamma = \text{undamping frequency [Hz]} \\ \Omega = \text{damping factor [Hz]} \end{cases}$$

After the Fourier transform, and further derivations, we get the position of the electron  $x(t)$ :

$$x(t) = \frac{-eE_0}{2m} \frac{\exp\{-j\omega_0 t\}}{\Omega^2 - \omega_0^2 - j2\Gamma\omega_0} + c.c.$$

#### 13.1.2 Total Medium Polarization

$$p(t) = -Nex(t) = \epsilon_0 \chi^{(1)} E(t)$$

Where  $\chi$  is the linear susceptibility, defined as:

$$\chi^{(1)}(\Omega, \omega_0) = \frac{Ne^2}{\epsilon_0 m} \frac{1}{\Omega^2 - \omega_0^2 - j2\Gamma\omega_0}$$

Since  $\vec{D} = \epsilon_0(1 + \chi^{(1)})\vec{E} = \epsilon_0\epsilon_r\vec{E}$  and  $\epsilon_r = n^2 \Rightarrow n = \sqrt{1 + \chi^{(1)}}$

The  $j2\Gamma\omega_0$  term is complex, this is the source of losses along the tx medium line.

If we are at a  $\omega_0 \neq \Omega \Rightarrow \chi^{(1)} \in \mathbb{R}$  in fiber we are away of this condition, but we have Rayleigh Scattering.

#### 13.1.3 Second Harmonic Generation

So where is non linearity? Dropping damping factor  $\Gamma$ , and Taylor-expanding the  $x(t, \delta)$  we get:

$$O(\delta^0) \rightarrow \frac{d^2x_0}{dt^2} + \Omega^2 x_0 = -\frac{e}{m} E(t)$$

$$O(\delta^1) \rightarrow \frac{d^2x_1}{dt^2} + \Omega^2 x_1 = -\Omega^2 x_0^2(t)$$

The  $x_0^2$  term is a  $\cos^2 \rightarrow$  it has a double frequency  $2\omega_0 \rightarrow$  it generates 2<sup>nd</sup> harmonic.

This is the process behind green laser ray generation (from infrared at 1000 nm).

In fibers we do not have second harmonics.

With WDM we get a non-linear effect of 4-wave mixing:  $2\omega_1, 2\omega_2, \omega_1 + \omega_2, \omega_1 - \omega_2$

#### 13.1.4 Non-Linear Medium Polarization

$$\vec{p}(\vec{e}) = \epsilon_0 \left( \chi^{(1)} \vec{e} + \chi^{(2)} : \vec{e}\vec{e} + \chi^{(3)} :: \vec{e}\vec{e}\vec{e} \right)$$

Fibers are made of Silica  $\rightarrow$  it is an Amorphous material, it guarantees Centro-Symmetric properties (changing the input sign implies linearly a change of the output signs). Being Centro-Symmetric we have that  $\vec{p}(\vec{e}) \neq \vec{p}(-\vec{e})$ . To get the equal sign the only way is to have  $\chi^{(2)} = 0$  so we don't have second harmonics, but instead we have third harmonic generation.

#### 13.1.5 Propagation in a Cubic Medium

Having  $\vec{E}(t) = E_0 \exp\{j\omega_0 t - j\beta z\}$  and  $\vec{e}(t) = \frac{1}{2} [E_0 \exp\{j\omega_0 t - j\beta z\} + c.c.]$  we can compute 3rd order of  $p$ :

$$p(t) = \frac{1}{8} \epsilon_0 \chi^{(3)} [3|E_0|^2 E_0 \exp\{j\omega_0 t - j\beta z\} + c.c.]$$

There is also another term  $\frac{1}{8} \epsilon_0 \chi^{(3)} E_0^3 \exp\{j3\omega_0 t - j3\beta z\}$ , but from the LP modes dispersion diagram we conclude this mode cannot propagate because of  $\beta(3\omega_0) \neq 3\beta(\omega_0)$ , **phase mismatching** problem!

### 13.1.6 Approximation of E

Trying an approximated analysis of  $E$ :

$$E \approx E_0(z)f(r, \phi)e^{-j\beta z}$$

where respectively  $E_0(z)$  is the complex field due to non-linearity,  $f(r, \phi)$  is the shape of the  $E$  and  $e^{-j\beta z}$  is the propagation factor. Plugging back this into Helmholtz equations and after some derivations, we get:

$$\frac{\partial E_0(z)}{\partial z} = -j \left( \frac{3\omega^2 \chi^{(3)}}{8\beta c_0^2} \frac{\int_s |F|^4 dx dy}{\int_s |F|^2 dx dy} \right) |E_0(z)|^2 E_0(z)$$

## 14 Lecture 5 Dec.

### 14.1 Non-Linearity

#### 14.1.1 Power flowing through fiber

We want to define some  $P$  such that  $|P(z)|^2$  has unit of measure of power [W], from the Poynting vector:

$$P(z) = \int_S (\vec{p} \cdot \vec{U}_z) dxdy = \frac{\beta}{2\omega\mu_0} |\hat{A}(z)|^2 \int_S |F|^2 dxdy \quad \Rightarrow \quad \frac{d\hat{p}}{dz} = -j\gamma |\hat{p}|^2 \hat{p}$$

Where  $\hat{A} = E_0$  of the previous lectures. We can consider  $\hat{p}(z) = \sqrt{P(z)}$  as the amplitude of the light.

#### 14.1.2 Non-linear coefficient $\gamma$

Defining the effective area (representing area occupied by the mode in the fiber)  $A_{eff}$  as:

$$A_{eff} = \frac{(\int_S |F|^2 dxdy)^2}{\int_S |F|^4 dxdy} \quad [\text{m}^2]$$

We can conclude the  $\gamma \propto A_{eff}^{-1}$ , what is the remaining part of the equation?

For similitude with Laplacian equation  $\nabla^2 \vec{E} + \frac{\omega^2}{c_0^2} n^2 \vec{E} = 0$ :

$$\gamma = \frac{\omega}{c_0} n_2 \frac{1}{A_{eff}} = \frac{2\pi}{\lambda_0} n_2 \frac{1}{A_{eff}}$$

This non-linear coefficient does not depend on any power, but only the electric field.  $n_2$  is the non-linear refractive index, depends on the material used.

#### 14.1.3 Pulse Propagation in a cubic medium

Defining complex amplitude of the wave  $\hat{a}(z) = \hat{p}(z) \exp\{-j\beta z\}$ , we have, depending on  $\omega$ :

$$\frac{\partial \hat{a}(\omega)}{\partial z} = -j\beta(\omega) \hat{a}(\omega) - j\gamma(\omega) |\hat{a}(\omega)|^2 \hat{a}(\omega)$$

#### 14.1.4 Non-Linear Schrödinger Equation (NLSE)

$$\frac{\partial a}{\partial z} = \left( -j\beta_0 a - \beta_1 \frac{\partial a}{\partial t} + j\frac{1}{2}\beta_2 \frac{\partial^2 a}{\partial t^2} \right) - j\gamma |a|^2 a$$

After some simplifications, we can neglect the first two terms in the parenthesis, such that:

$$\frac{\partial a}{\partial z} = j\frac{1}{2}\beta_2 \frac{\partial^2 a}{\partial t^2} - j\gamma |a|^2 a$$

What is the cause of non-linearity (second term)?

### 14.2 Non-Linearity Effects

#### 14.2.1 Self-Phase Modulation

We can further simplify: if  $\beta_2 \approx 0$  (with Zero Dispersion Fibers)  $\Rightarrow \frac{\partial |a|^2}{\partial z} = 0 \Rightarrow |a(z, t)|^2 = |a(0, t)|^2 = P(t)$   
Having  $a(z, t) = a(0, t) \exp\{-j\gamma P(t)z\}$ :

$$\frac{\partial a}{\partial z} = -j[\gamma P(t)] a(z, t)$$

#### 14.2.2 Chirping

If  $\gamma = 0$  we can neglect the term. Chirping is a problem for WDM, we allow a max frequency delta of 2.7 GHz. Basically we have a phase term  $\varphi(t) = -\gamma P(t)z$  which causes frequency derivation as  $f(t) = \frac{1}{2\pi} \frac{d\varphi(t)}{dt} = \frac{-\gamma}{2\pi} \frac{dP}{dt} z$

### 14.3 Effect tradeoff parameter over length

The shorter length between these ( $L_D, L_{NL}, L_{eff}$ ) determines which type of phenomenon to consider.



### 14.3.1 Non-Linear Length

Distance in which there are non-linear phenomena.

$$L_{NL} = \frac{1}{\gamma P_{\max}}$$

### 14.3.2 Effective Length (Attenuation)

$$L_{eff} = \frac{1 - \exp\{-\alpha z\}}{\alpha}$$

For  $z \rightarrow 0 \Rightarrow L_{eff} \approx z$ , while for  $z \rightarrow \infty \Rightarrow L_{eff} \rightarrow \frac{1}{\alpha}$ , typically 22km.

### 14.3.3 Dispersion Length

It represents the Chromatic Dispersion effect.

$$L_D = \frac{T_0^2}{|\beta_2|}$$

## 14.4 Solitons

Pulses that propagate without any distortion:

$$a(z, t) = \sqrt{P_0} \cdot \operatorname{sech} \left( \sqrt{\frac{\gamma P_0}{|\beta_2|}} t \right) \exp \left( -j \frac{\gamma P_0}{2} z \right)$$

Note that the sech term does not depend on distance, while the exp term doesn't depend on time. Only problem here is that we don't consider attenuation.